

2. Heat Risk Assessment and Spatial Analysis

Extreme heat does not affect all parts of a city equally. Variations in land surface temperature, vegetation cover, moisture conditions, built-up intensity, and population concentration create significant spatial differences in heat risk across the urban area. For a city-level Heat Action Plan, it is therefore essential to identify where heat stress is highest and where larger numbers of people are likely to be affected.

To support planning in Shahjahanpur, a spatial heat risk assessment was carried out using remotely sensed environmental indicators, building density, and population distribution data. In addition, land use, labour chowk locations, and Urban Primary Health Centre (UPHC) locations were used as supporting layers to interpret practical implications for mitigation, worker protection, and emergency response.

The objectives of this chapter are to:

- Identify areas experiencing higher physical heat exposure
- Assess spatial concentration of the sensitive population and built-up conditions
- Generate a composite heat risk index
- Identify priority heat-risk areas for targeted intervention under the heat action plan

The results of this chapter provide the spatial evidence base for prioritising actions related to urban cooling, public awareness, worker safety, and health Preparedness.

Methodological Innovation and Administrative Value

Traditionally, local heat response strategies have relied on district-wide meteorological alerts, resulting in blanket, reactive advisories that treat the entire urban area uniformly. The unique contribution of this chapter is the development of a high-resolution, spatially explicit modelling framework that bridges advanced remote sensing analytics with local disaster management protocols.

By systematically processing mid-morning thermal and ecological data from Landsat 8/9 and intersecting it with high-resolution population grids (GHSL) and building footprint geometries, this assessment moves beyond basic temperature mapping. It quantifies the exact spatial intersection where severe physical heat exposure compounds with high socio-spatial vulnerability.

For the Uttar Pradesh State Disaster Management Authority (UPSDMA) and local district administrators, this methodological innovation provides a transformative administrative tool. It shifts heat wave management from a generalized public health advisory to an **evidence-based, micro-targeted resource allocation strategy**. By isolating the "central urban heat corridor," this analysis enables the District Administration to precisely deploy limited resources, such as emergency water tankers, temporary shading, and cooling centres, to the highest relative risk zones. Ultimately, this spatial intelligence forms the empirical foundation for all department-wise Standard Operating Procedures (SOPs) executed during Red and Orange alerts.

Conceptual Framework

The present assessment adopts a simplified operational framework:

$$\text{Heat Risk} = \text{Exposure} \times \text{Sensitivity}$$

This approach effectively identifies the city's central urban belts where extreme heat and dense human settlement overlap. However, a holistic climate resilience approach requires evaluating Heat Vulnerability. In addition to Exposure and Sensitivity, a true Heat Vulnerability Index incorporates a third critical dimension: Adaptive Capacity. **Adaptive capacity** refers to the resources, infrastructure, and institutional readiness available to a community to cope with and adapt to severe heat stress.

Because spatially explicit adaptive-capacity indicators such as water supply reliability, electricity access, healthcare accessibility, cooling centres, and institutional preparedness were not available in a form suitable for citywide quantitative integration, this approach was adopted. Accordingly, the present chapter presents a Heat Risk Index rather than a full Heat Vulnerability Index, serving as an actionable first step.

Exposure refers to the physical intensity and distribution of heat-related environmental conditions across the city. In this assessment, exposure was represented by:

- Mean Land Surface Temperature (LST)
- vegetation deficit, represented through inverse NDVI
- surface moisture deficit, represented through inverse MNDWI

Sensitivity refers to the concentration of population and built-up conditions that increase the likely burden of heat stress. In this assessment, sensitivity was represented by:

- population distribution
- building density

Seasonal Rationale

April and May were selected to represent the peak pre-monsoon summer season in Shahjahanpur. During this period, solar heating is intense and monsoonal rainfall has not yet substantially modified urban thermal and moisture conditions. This seasonal window therefore provides an appropriate basis for mapping peak heat stress across the city.

Data Sources

The heat risk assessment was based on the following datasets:

Table 1: Data Sources and Specifications for Heat Risk Assessment

Data	Source	Period	Resolution	Purpose
Landsat 8/9 Collection 2 Level 2	USGS/NASA via Google Earth Engine	April-May 2025	30 m	Mean LST, NDVI, MNDWI
Microsoft Building Footprints	Open building footprint dataset	Current reference	Vector to raster	Building density
GHSL P2023A GHS_POP	JRC / GHSL	2025	100 m	Population distribution
AMRUT Scheme land use map	AMRUT planning dataset	Current	Vector	Land use interpretation
Labour chowk locations	Local/municipal data	2025	Point	Occupational heat exposure context
UPHC locations	Municipal/health data	2025	Point	Health response context

Methodology

The analytical procedure involved the following steps:

1. Landsat 8 and Landsat 9 imagery was filtered for Shahjahanpur city and restricted to April-May 2025.
2. Cloud-masked Landsat imagery was processed to derive Mean Land Surface Temperature, NDVI, and MNDWI.
3. Microsoft Building Footprints were converted into a raster building density layer representing building footprint area per 30 m grid cell.
4. The GHSL 2025 raster was used to map population count per 100 m grid cell.
5. All variables used in the composite analysis were normalised to a common scale of 0 to 1 using min-max normalisation.
6. An Exposure Index was generated from Mean LST, inverse NDVI, and inverse MNDWI.
7. A Sensitivity Index was generated from population and building density.
8. The final Heat Risk Index was calculated by combining exposure and sensitivity.
9. The resulting spatial patterns were interpreted with reference to land use, labour chowks, and UPHC locations.

Bimonthly Temperature Trend (2013-2025)

To provide temporal context to the spatial heat risk assessment, a bimonthly temperature trend analysis was carried out for the period 2013-2025 using Landsat-derived mean and extreme land

surface temperatures. Academically, this longitudinal time-series data confirms that the severe thermal hotspots identified in the spatial maps are not single-year anomalies, but rather the result of a compounding, decade-long climatic shift. The trend graph demonstrates progressively recurring seasonal peaks (above 53 °C) during the pre-monsoon summer months.

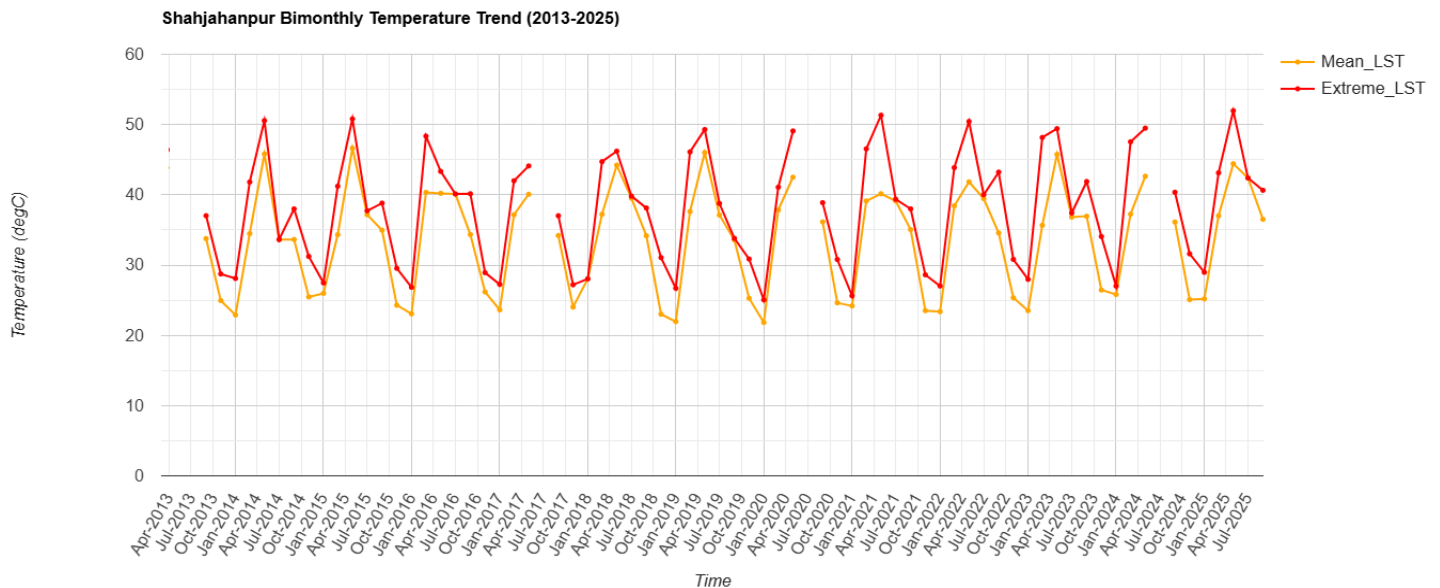


Figure 2: Shahjahanpur Bimonthly Temperature Trend (2013–2025)

For the UPSDMA and district policymakers, this 12-year empirical evidence is highly actionable: it proves that heat waves in Shahjahanpur can no longer be treated as ad-hoc, seasonal emergencies. Instead, it mandates that heat mitigation must be institutionalised as a permanent administrative priority, requiring dedicated annual budget allocations and year-round infrastructural resilience planning.

Indicator formulation

$$\text{Exposure} = 0.50 (\text{Mean LST}) + 0.30 (1 - \text{NDVI}) + 0.20 (1 - \text{MNDWI})$$

$$\text{Sensitivity} = 0.60 (\text{Population}) + 0.40 (\text{Building Density})$$

$$\text{Heat Risk} = \text{Exposure} \times \text{Sensitivity}$$

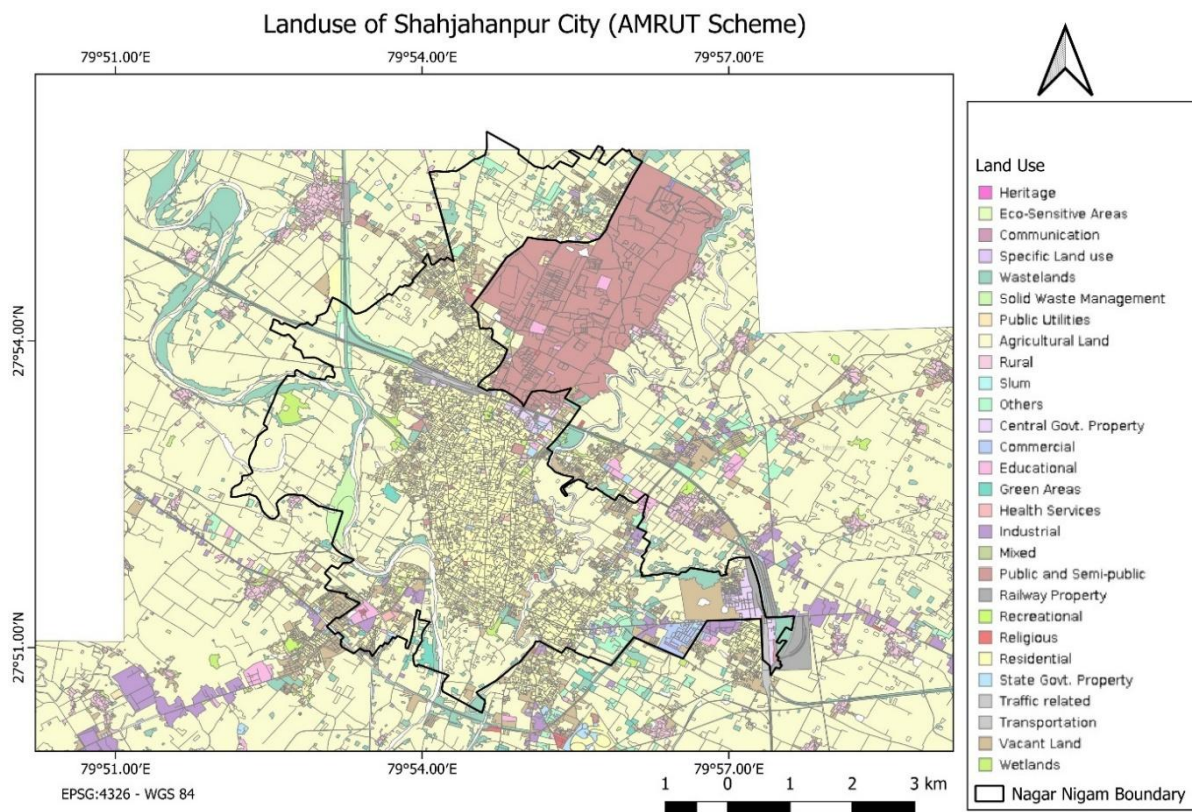
Analytical Weighting Rationale The mathematical weighting of these indices was deliberately structured to reflect both physical climatology and disaster management priorities. In the Exposure equation, Mean LST is assigned the highest weight (50%) as it represents the primary thermal hazard, while vegetation (30%) and moisture deficits (20%) act as secondary compounding modifiers. More importantly, in the Sensitivity equation, human population is weighted at 60% compared to building density at 40%. For the District Administration, this weighting is a critical policy choice: it ensures that the final Heat Risk Index mathematically prioritizes the preservation of human life over the mere physical heating of infrastructure, ensuring that emergency resources are directed strictly to where the most people are at risk.

Indicator units

- Mean LST: degree Celsius (°C)
- NDVI: unitless
- MNDWI: unitless
- Building density: building footprint area per 30 m pixel (m²)
- Population: people per 100 m grid cell

Land Use Context

The AMRUT land use map provides an important planning context for interpreting heat conditions in Shahjahanpur. Although land use was not directly included as a numerical variable in the composite index, it helps explain the spatial pattern of built-up areas, transport corridors, public and semi-public land, green areas, open land, and agricultural land.



Map 1. Land Use of Shahjahanpur City (AMRUT Scheme).

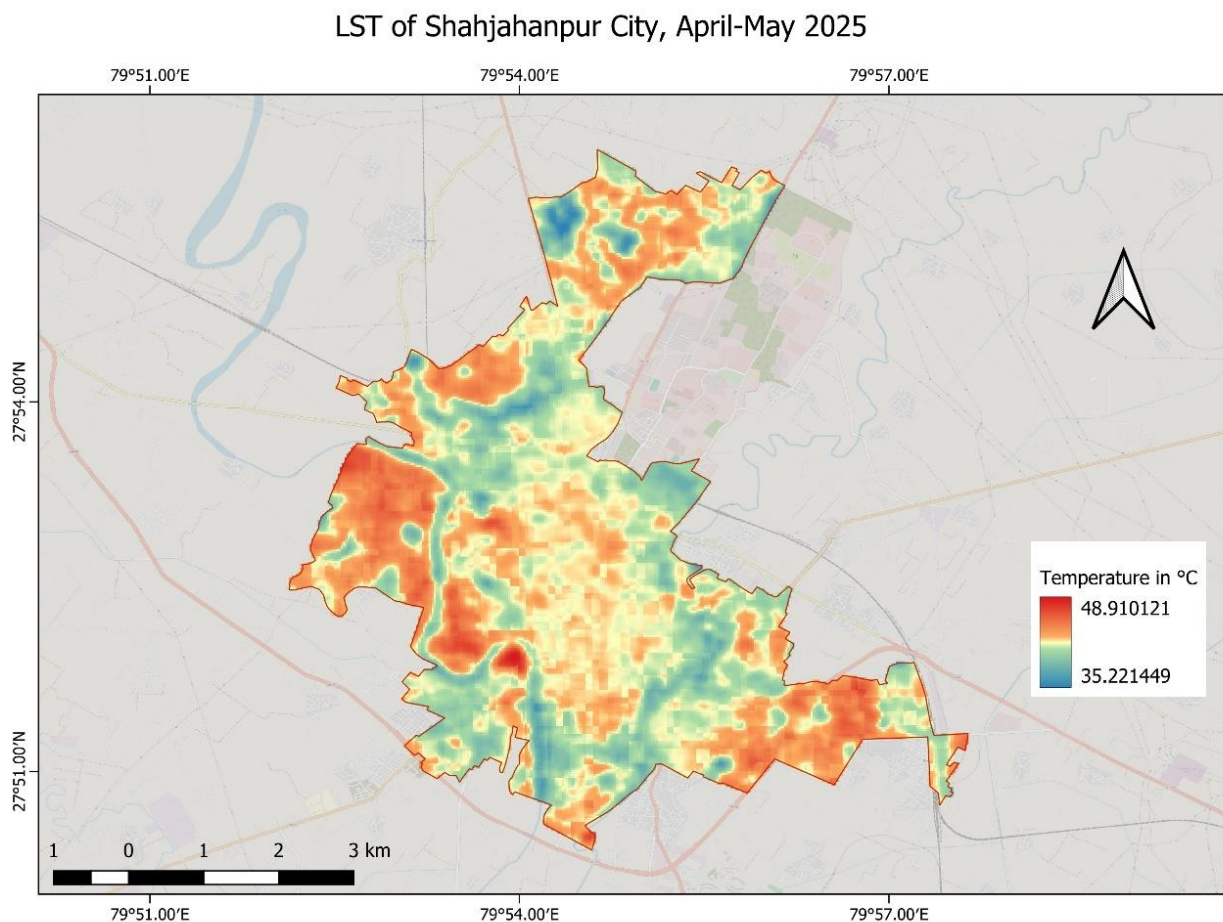
ENVIRONMENTAL INDICATORS OF HEAT EXPOSURE

To effectively manage urban heat, we must first quantify the physical hazard generated by the city's built landscape. The environmental indicators of heat exposure—comprising Land Surface Temperature, vegetation deficits, and surface moisture—combine to map the baseline climatological risk across Shahjahanpur.

Mean Land Surface Temperature

The Mean LST map shows the average spatial pattern of surface temperature during April-May 2025. Higher LST values indicate areas with stronger heat accumulation, often associated with impervious surfaces, built-up land, and reduced vegetation or moisture cover.

Translating Thermal Dynamics into Targeted Mitigation: The spatial correlation between the AMRUT land use (Map 1) and Mean LST (Map 2) demonstrates a fundamental thermal dynamic: the city's highly impervious commercial and dense residential core acts as a primary heat receptor. With surface temperatures peaking from 45°C to 48°C in these zones (in the western periphery of city the river bed also gets the temperature around 48.9° but it cools down at night due to low absorption of radiation), the maps visually prove how dense urban materials absorb and trap significantly more heat than peripheral agricultural landscapes.

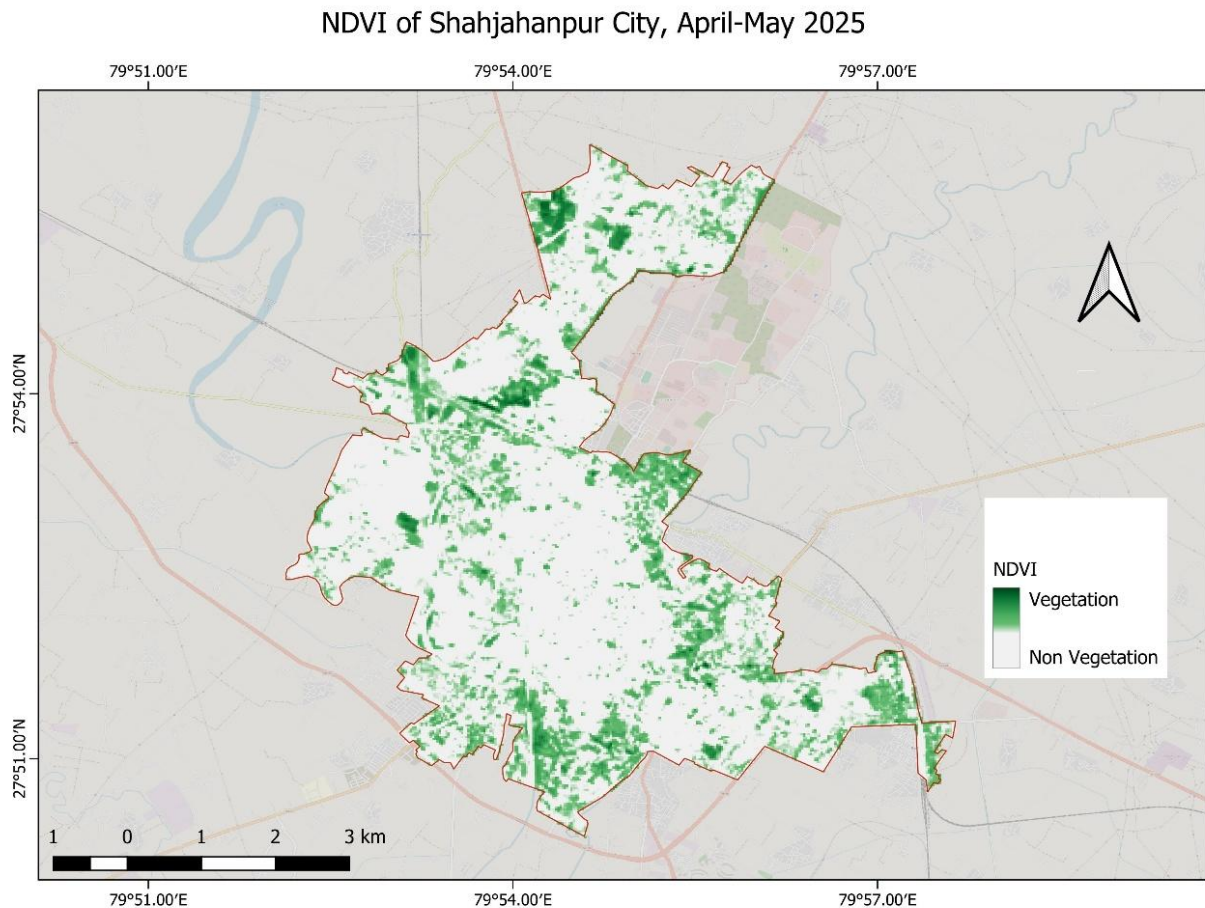


Map 2. Mean Land Surface Temperature of Shahjahanpur City, April-May 2025

For the Uttar Pradesh State Disaster Management Authority (UPSDMA) and Nagar Nigam, this spatial evidence shifts heat mitigation from blanket, city-wide advisories to micro-targeted interventions. The dark red and orange thermal hotspots on Map 2 dictate the exact municipal locations where cool-roofing protocols, reflective paints, and peak-hour water sprinkling must be prioritized and enforced

Vegetation Cover (NDVI)

The NDVI map shows the distribution of vegetation cover across the city. NDVI ranged from -0.64 to 0.70 , indicating a wide gradient from built-up/barren surfaces, Agriculture to dense vegetation. Higher NDVI values indicate denser vegetation, which contributes to urban cooling through shading and evapotranspiration. Lower NDVI values indicate reduced green cover and, therefore, greater susceptibility to heat stress. In Map 3, we can clearly see that the middle of the north-to-south parts of the city, which include transport networks and industrial clusters, don't have enough tree cover. In contrast, the city's periphery has agricultural areas bordering both rivers.

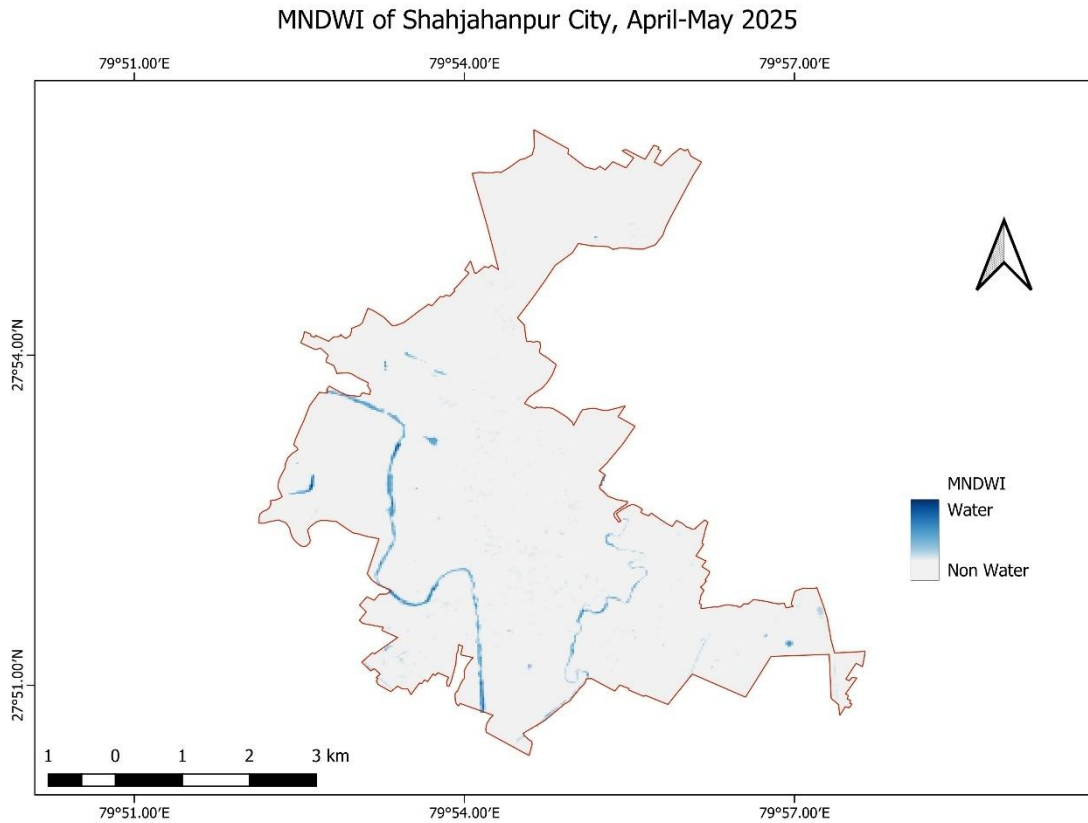


Map 3. NDVI of Shahjahanpur City, April-May 2025.

Surface Moisture Conditions (MNDWI)

The MNDWI map represents surface moisture and water-related landscape features. MNDWI ranged from -0.5 to 0.35 , where positive values indicate surface water presence. MNDWI was preferred over NDWI because it performs better in urban areas by reducing confusion with built-up surfaces. In the exposure model, lower MNDWI values indicate lower moisture availability and therefore contribute to higher heat exposure.

Map 4 clearly shows that the city and its settlements have emerged between the rivers Garrah (west) and Khannaut (east). These two major water features moderate the city's temperature to a certain degree, along with a few ponds. There are a few fishing ponds in the north of city. Overall, the impact of these waterbodies is insufficient to counteract daytime LST in the city due to anthropogenic heat.



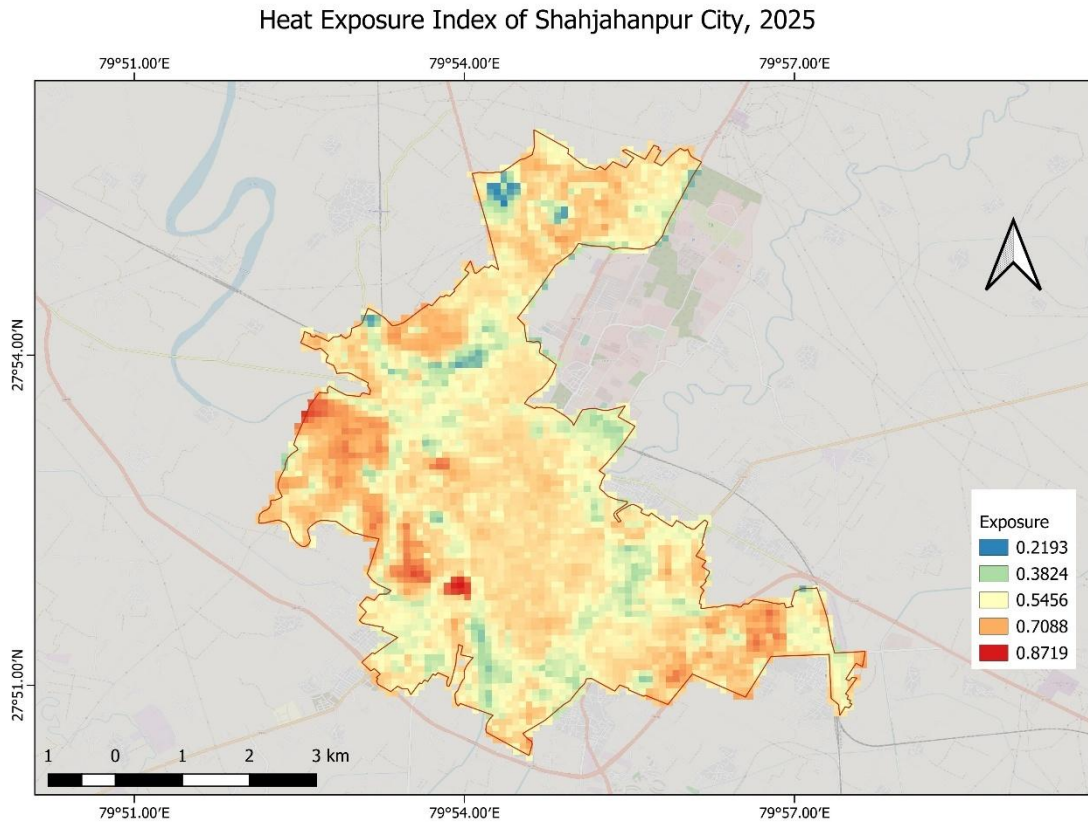
Map 4. MNDWI of Shahjahanpur City, April-May 2025.

Quantifying Ecological Deficits for Long-Term Resilience

Maps 3 (NDVI) and 4 (MNDWI) reveal a severe ecological fragmentation within the urban core. From a spatial modelling perspective, the deficit in green-blue infrastructure severely restricts natural evapotranspirative cooling, acting as a multiplier for thermal stress. By mathematically integrating these inverse ecological values with the LST data, the composite Heat Exposure Index (Map 5) is generated. Administratively, this composite map provides the Forest Department and the District Magistrate with empirical coordinates for prioritizing climate adaptation. The spatial polygons exhibiting severe ecological deficits serve as the exact, evidence-based locations for deploying Miyawaki forests and restoring localized water bodies under state resilience schemes.

Heat Exposure Index: Synthesizing the Physical Thermal Hazard

The Heat Exposure Index (Map 5) is a mathematically weighted composite of the city's physical thermal hazard. By integrating Mean LST (weighted at 50%), vegetation deficit (30%), and surface moisture deficit (20%), this model quantifies exactly how the absence of natural cooling mechanisms compounds surface heat. The resulting map shows that physical exposure is heavily concentrated in the built-up and environmentally stressed central sectors of the city.



Map 5. Heat Exposure Index of Shahjahanpur City, 2025.

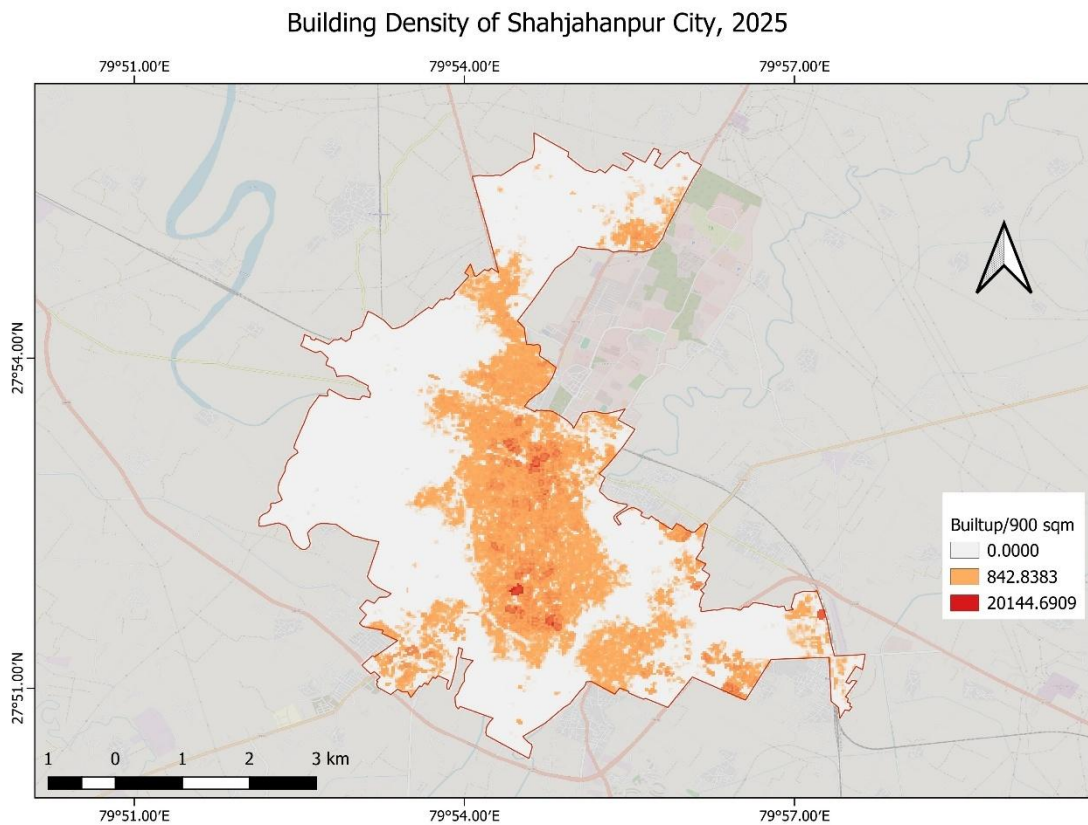
From an administrative perspective, this map isolates the exact geographic zones where the urban environment itself has become inherently hostile. For the UPSDMA, Nagar Nigam, and the Forest Department, the dark red exposure hotspots provide the empirical baseline for structural climate adaptation. These specific areas demand long-term interventions that go beyond temporary seasonal relief, such as mandating cool-roofing in commercial zones, aggressively deploying Miyawaki urban afforestation in empty plots in the city core, and restoring localized water bodies to reconstruct the city's lost ecological buffering capacity.

SENSITIVITY INDICATORS

While the environmental indicators (LST, NDVI, MNDWI) map the physical hazard of extreme heat, the spatial risk equation is incomplete without understanding the receptors of that hazard. Sensitivity refers to the spatial concentration of human populations and the compact built-up conditions that amplify thermal stress. In this spatial assessment, the Sensitivity metric is driven by two heavily weighted variables: human population distribution (60%) and building footprint density (40%). For the District Administration and UPSDMA, mapping these sensitivity indicators represents a critical shift from merely monitoring the weather to actively monitoring human vulnerability. It answers the most urgent administrative question during a heatwave alert: Where are the maximum number of people trapped in the most congested environments, and where should emergency relief — such as water tankers and power restoration — be prioritised?

Building Density: Mapping Structural Heat Retention

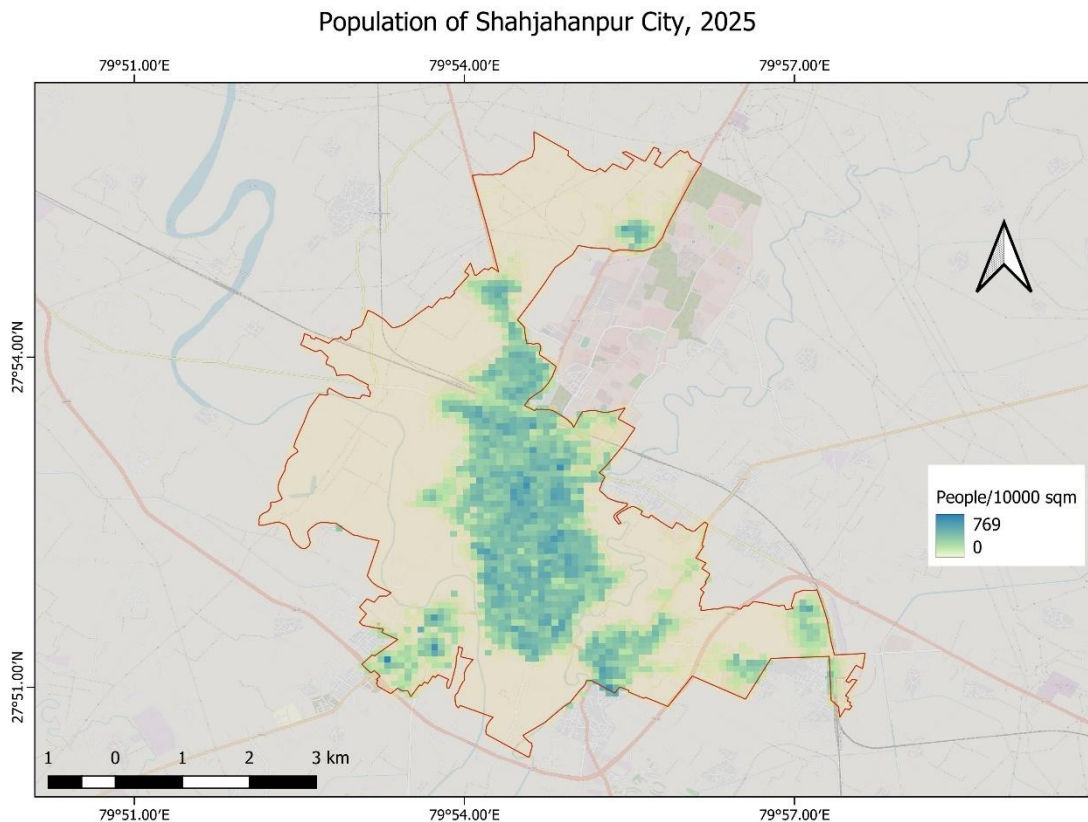
The building density map (Map 6) highlights the physical concentration of built structures across Shahjahanpur. From a spatial-analytical perspective, areas with a highly dense built-up fabric, indicated by the dark red and orange clusters in the city center, suffer from severe heat retention, restricted natural airflow, and an exacerbated urban heat island effect. Because tightly packed concrete structures and tin roofs absorb and trap heat day and night, these zones act as structural amplifiers of thermal stress. For the District Administration and Nagar Nigam, this map serves as a structural intervention guide. It visually isolates the exact congested areas where homes have the poorest ventilation. These are the priority zones for enforcing cool roofing guidelines, deploying temporary shading at public squares, and ensuring the Fire Department is pre-positioned for heat-induced emergencies in narrow, highly-built corridors.



Map 6. Building Density of Shahjahanpur City, 2025.

Population Distribution: Identifying Human Exposure Zones

The population map (Map 7) utilizes the GHSL 2025 spatial layer to project the distribution of residents, shifting the assessment from physical infrastructure to human vulnerability. The map visually highlights severe human concentration, with the dark blue-green zones indicating densities reaching up to 769 people per 10,000 square meters. Academically, this high spatial concentration indicates exactly where a heatwave will have the maximum statistical probability of causing mass morbidity.

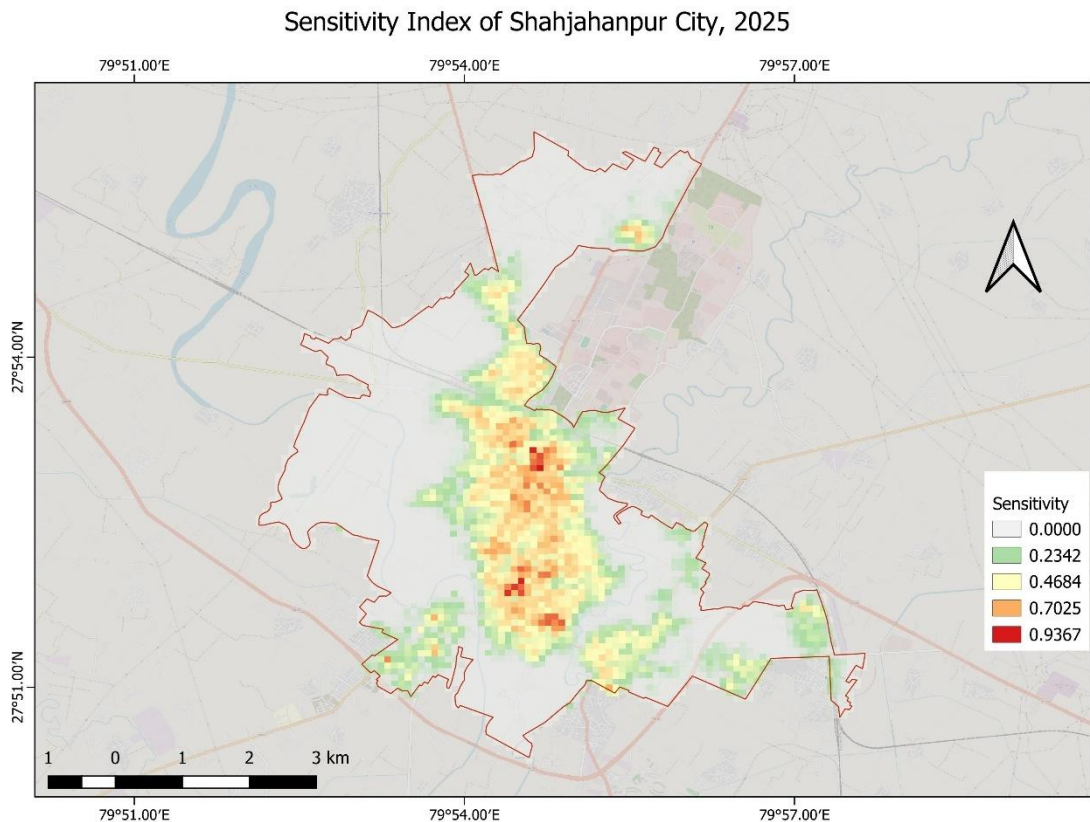


Map 7. Population of Shahjahanpur City, 2025.

For the DDMA, Health Department, and Jal Nigam, these dark blue-green demographic hotspots dictate immediate operational logistics. During an Orange or Red alert, these specific densely populated wards require the highest volume of mobile water tanker deployment, priority electricity feeder maintenance to prevent mass cooling failures, and the concentrated deployment of ASHA and Anganwadi workers for proactive heat-illness surveillance.

Sensitivity Index: Where Vulnerable Populations Live

Extreme heat only becomes a disaster when it affects people. The Sensitivity Index combines population and building density to show us exactly where the most people are living in closely packed conditions. Higher sensitivity values indicate areas where dense settlement and compact built form overlap, thereby increasing the likely burden of heat stress on residents. These highly sensitive zones often include narrow streets and informal settlements where homes have poor ventilation, limited water storage, and limited access to cooling devices. During an extreme heatwave, these densely populated neighbourhoods will require the most immediate administrative relief, such as uninterrupted drinking water supply, targeted electricity maintenance, and the opening of nearby community cooling centres.

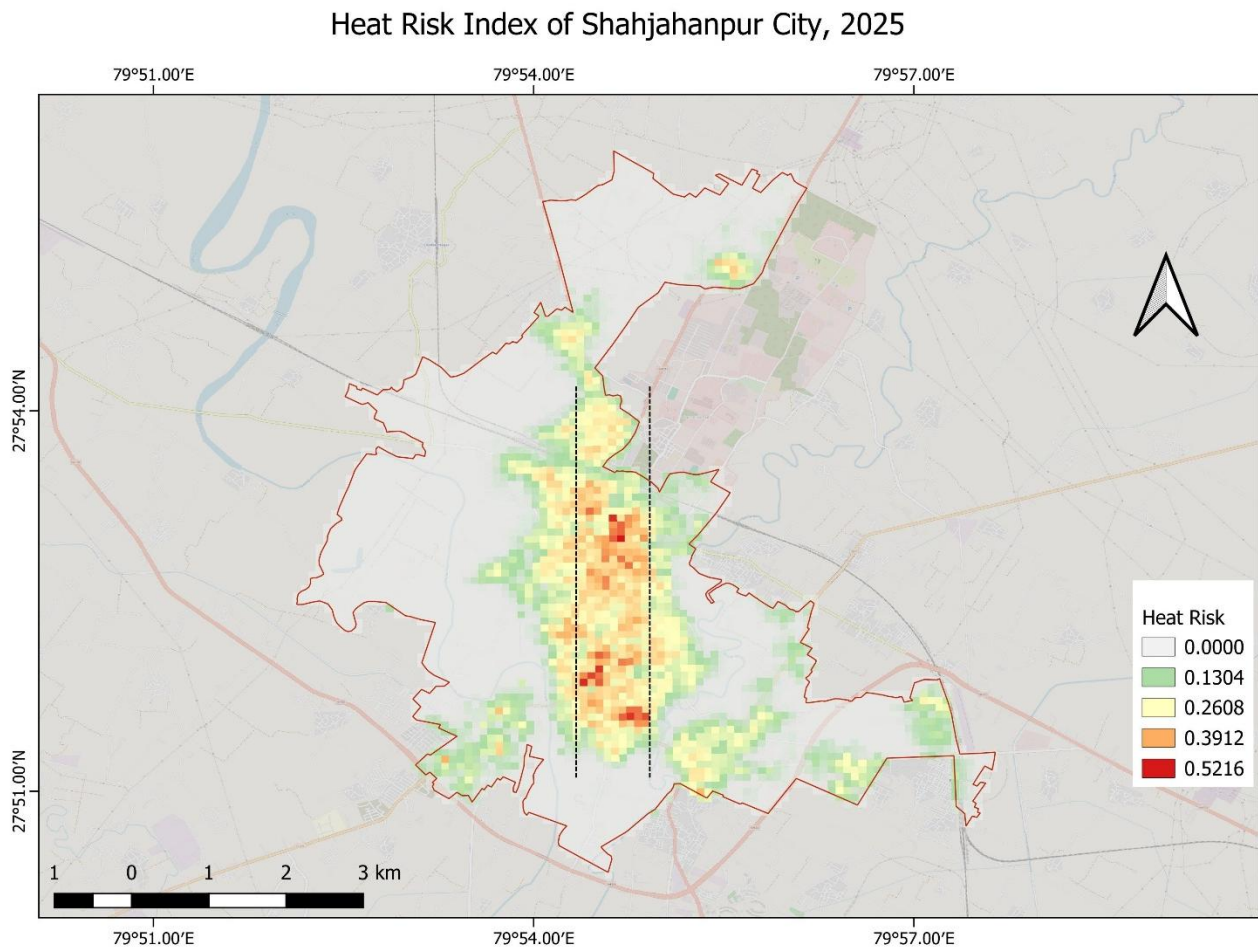


Map 8. Sensitivity Index of Shahjahanpur City, 2025.

COMPOSITE HEAT RISK

Heat Risk Index: Identifying the High-Risk Heat Corridor for Administrative Action

Heat Risk Index combines physical heat exposure with population sensitivity to identify the parts of Shahjahanpur where extreme heat and human concentration overlap. The map identifies a clear "central urban heat corridor" extending from the north-central to the south-central parts of Shahjahanpur. This map provides the exact spatial evidence the District Administration needs to prioritize their response. When the IMD issues an orange or red alert, district resources—such as mobile water tankers, shaded shelters, and targeted public awareness campaigns—must be deployed first to this specific high-risk corridor to prevent avoidable health emergencies.



Map 9. Heat Risk Index of Shahjahanpur City, 2025.

The Final Heat Risk Picture: The composite Heat Risk Index map brings all the data together into one actionable visual. The legend uses a scale from 0 to 0.52, where the deep red color denotes the absolute highest risk. The map clearly tells us that the greatest heat risk is not scattered randomly; it forms a distinct, continuous "red corridor" running straight through the north-central to south-central parts of the city. For district planners and emergency responders, this legend and map provide a direct, color-coded boundary. It tells the administration exactly which specific neighborhoods require the first dispatch of water tankers, the most urgent electricity maintenance, and the immediate opening of cooling shelters during a severe heatwave.

Visual Validation of the Spatial Model: To validate this spatial risk map for immediate administrative use, seven critical hotspots were ground-truthed using high-resolution Google Earth Pro snapshots. These images provide direct visual proof of the severe urban conditions—such as closely spaced roofing and a severe lack of natural shade—driving the high-risk scores. For the District Administration, these visually verified hotspots serve as the undeniable 'ground zero' for executing the emergency interventions of the Shahjahanpur Heat Action Plan.

Heat Risk Hotspots of Shahjahanpur City, 2025

Satellite-based visual interpretation of seven selected urban locations

This figure connects the city heat risk map with neighborhood satellite views to show how dense construction, paved roads, and limited vegetation shape localized urban heat hotspots across Shahjahanpur.

City-Scale Heat Pattern

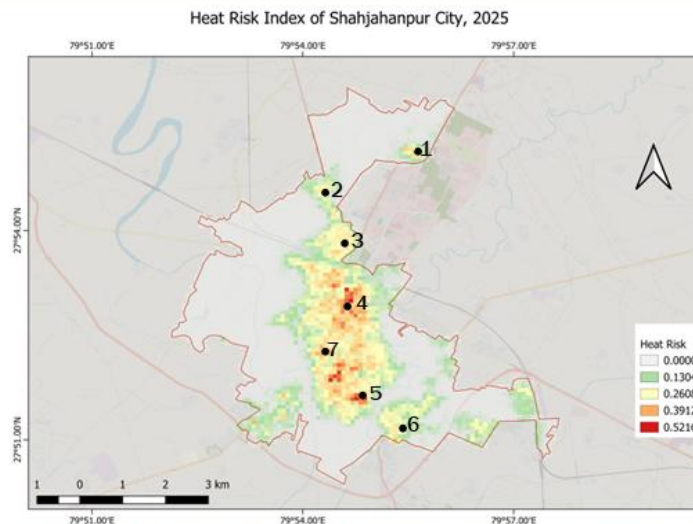


Figure 1. Heat Risk Index of Shahjahanpur City, 2025. Warmer colors indicate stronger heat-risk concentrations.

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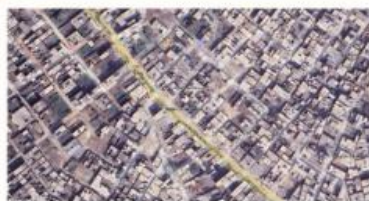
Hotspots
linked to
satellite
evidence

Heat
story



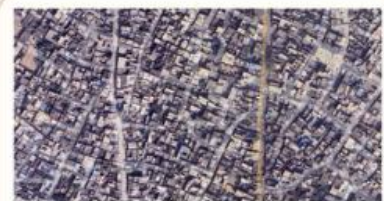
1 North Shahjahanpur

Localized heat pocket shaped by clustered built surfaces and limited vegetation.



2 Satwan Bujurg

Compact settlement pattern likely increasing surface heat retention and reducing airflow.



3 Ramnagar Colony

Mixed residential density with paved lanes, concrete roofs, and low tree cover.



4 Sadar Bazar

Major urban hotspot driven by dense construction and intense commercial activity.



5 Keruganj

Highly compact built-up area with strong rooftop and road heat absorption.



6 Dalel Gunj

Moderate hotspot shaped by dense blocks, hard surfaces, and limited shaded space.



7 Anjan Chowk

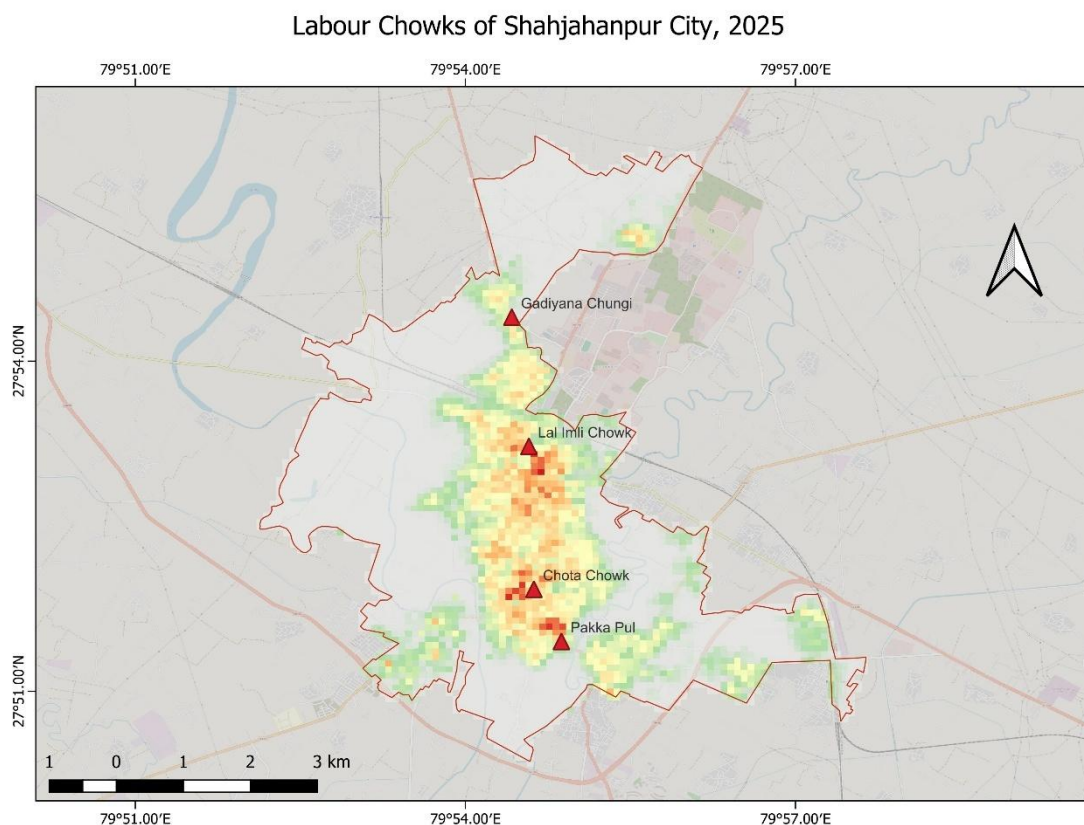
Road-junction hotspot where dense rooftop traffic surfaces, and low greenery combine

- Dense buildings reduce airflow and trap heat
- Concrete, asphalt, and rooftops store solar energy
- Low tree cover reduces shade and evapotranspiration
- Commercial and road-junction zones intensify heat stress

OCCUPATIONAL EXPOSURE & HEALTH RESPONSE CONTEXT

Labour Chowks

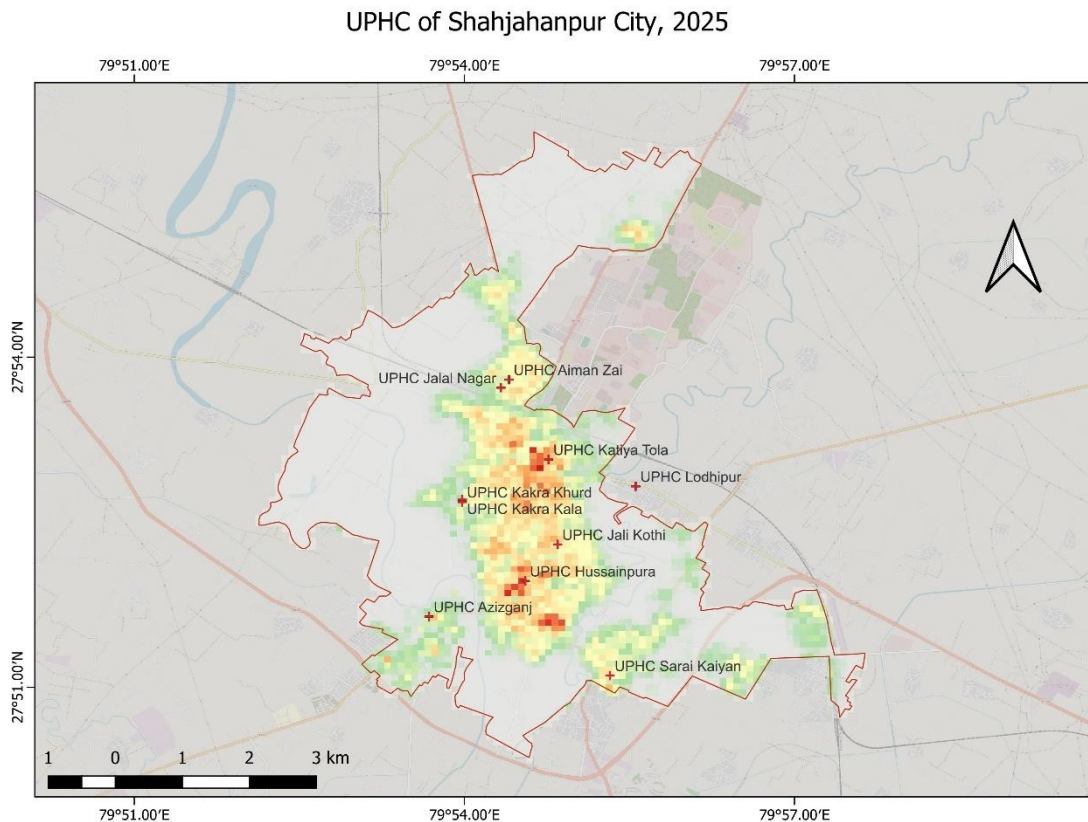
The labour chowk map identifies locations where outdoor and informal workers gather, often during daytime hours when thermal stress is high. Several labour chowks fall within moderate to high heat-risk zones, highlighting the need for worker-focused interventions such as shaded waiting areas, safe drinking water, adjusted work timings, and targeted heat advisories.



Map 10. Labour Chowks of Shahjahanpur City, 2025.

Urban Primary Health Centres (UPHCs)

The UPHC map was examined as a supporting health Preparedness layer. Several UPHCs are located within or near the principal heat-risk belt, indicating their importance in local heat-health response. However, the present map only shows facility locations and does not evaluate accessibility, staffing, service adequacy, or emergency capacity. It should therefore be interpreted as a contextual planning layer rather than a quantified adaptive-capacity indicator.



Map 11. UPHCs of Shahjahanpur City, 2025.

Overlaying the Labour Chowks (Map 10) onto the high-risk Heat Risk Index (Map 9) highlights a severe occupational vulnerability: daily-wage and informal workers are forced to congregate and labor in the city's most thermally hostile environments. While the geographical distribution of UPHCs (Map 11) shows proximity to these high-risk belts, spatial proximity alone does not equate to adaptive capacity. For the UPSDMA, Health, and Labour Departments, this spatial intersection acts as an operational trigger. It necessitates strict administrative enforcement of work-hour modifications (restricting heavy work between 12 PM and 4 PM during Red alerts) and mandates that these specific, high-risk local clinics are pre-stocked with ORS, IV fluids, and dedicated heatstroke beds prior to the summer peak.

Spatial Interpretation of Priority Heat-Risk Areas

The overall analysis indicates that heat risk in Shahjahanpur is concentrated in a central urban corridor extending broadly from the north-central city toward the south-central city. This corridor combines:

- Relatively elevated surface temperatures
- Low vegetation and moisture conditions
- Higher population concentration
- Greater built-up intensity

Peripheral areas generally display lower heat risk because of lower population concentration, lower building density, and in some cases relatively better vegetation and moisture conditions. The land use map supports this interpretation by showing the concentration of urban development in the city

core. The labour chowk and UPHC maps further show that several important worker congregation points and health service nodes are located close to or within these higher-risk zones.

Limitations and Scope for Future Refinement

The present assessment is a spatial heat risk analysis rather than a full heat vulnerability assessment. While it captures physical exposure and urban sensitivity, it does not yet include a quantitatively mapped adaptive-capacity component. Future assessments should expand on both the Sensitivity and Adaptive Capacity dimensions.

Refining Sensitivity (Vulnerable Populations): Currently, sensitivity is measured at a general level using total population and building density. In the future, the Sensitivity Index should include more specific demographic variables whenever high-resolution data becomes available. Important highly-vulnerable micro-populations that should be incorporated into the sensitivity metric include:

- Informal settlements, Homeless population
- Outdoor occupational groups at neighbourhood scale
- Elderly population
- Children below six years
- Pregnant women

Incorporating Adaptive Capacity: Furthermore, to transition to a complete Heat Vulnerability Index, future refinements should map indicators of adaptive capacity, which include:

- Households lacking reliable water access
- Households lacking reliable electricity access
- Healthcare accessibility and facility capacity
- Cooling centres and community shelters

Incorporation of such variables would enable development of a more comprehensive Heat Vulnerability Index for Shahjahanpur.

Conclusion: From Spatial Data to Administrative Action

The spatial analysis demonstrates that heat risk in Shahjahanpur is not randomly distributed, but heavily concentrated in a central urban corridor shaped by high thermal retention, severe ecological deficits, and dense human congestion. For the District Administration, the Composite Heat Risk Index and supporting infrastructural maps represent a shift from reactive disaster management to proactive, precision governance. However, identifying these spatial hotspots is only the first step. To operationalize the Heat Action Plan, the administration must understand how this mapped physical exposure translates into socio-economic vulnerability and systemic infrastructure stress. The following chapter transitions from spatial hazard mapping to human impact, detailing the exact demographic groups and critical urban systems that require immediate institutional protection within these high-risk zones.